# FBISE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

# CHEMISTRY

Past Paper 2023 • Paper Code: 2HA-I 2309 — B(4091) / D(8091)

Retyped faithfully from the original board paper — same questions, no changes.

## VERSION B

Note: CHEMISTRY HSSC-II. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 4091. Fill the relevant bubble against each question according to curriculum:

Q1. Which of the following has highest ionization energy?

(A) Lithium
(B) Boron
(C) Oxygen
(D) Nitrogen

Q2. Which of the following oxides is acidic?

(A) $Na_2O$
(B) $KO_2$
(C) MgO
(D) $NO_2$

Q3. $Pb^{+4}$ compounds are good:

(A) Reducing agents
(B) Oxidizing agents
(C) Dehydrating agents
(D) Hydrating agents

Q4. Carboxylic acids can be prepared by the reactions of Grignard reagent with:

(A) Oxygen
(B) Nitrogen
(C) Carbon dioxide
(D) Formaldehyde

Q5. To what category of molecules do enzymes belong?

(A) Carbohydrates
(B) Proteins
(C) Lipids
(D) Fats

Q6. Peroxyacetyl Nitrate (PAN) is an irritant to human beings and it affects:

(A) Eyes
(B) Nose
(C) Stomach
(D) Skin

Q7. Chemically nail polish remover is:

(A) Acetone
(B) Acetic acid
(C) Ethanol
(D) Ether

Q8. Which of the following electronic transitions requires highest amount of energy when a compound is interacted with ultraviolet radiations?

(A) n→π*
(B) π→π*
(C) δ→δ*
(D) n→δ*

Q9. Height of peak in mass spectrum shows:

(A) Number of isotopes
(B) Mass number
(C) Relative abundance of isotopes
(D) Number of protons

Q10. Which of the following will NOT give iodoform test?

(A) Formaldehyde
(B) Acetaldehyde
(C) Acetone
(D) Ethanol

Q11. Oxidation of 2-Propanol gives:

(A) Acetaldehyde
(B) Formaldehyde
(C) Acetone
(D) Acetic acid

Q12. Nitration of phenol takes place at ____ than benzene.

(A) Slower rate
(B) Faster rate
(C) Equal rate
(D) Moderate rate

Q13. Reaction of alcohols with sodium metal gives:

(A) Helium gas
(B) Hydrogen gas
(C) Carbon dioxide
(D) Carbon monoxide

Q14. Geometrical isomerism is shown by:

(A) Lactic acid
(B) Acetic acid
(C) 1-Butene
(D) 2-Butene

Q15. Which of the following alkynes is acidic in nature?

(A) 1-Butyne
(B) 2-Butyne
(C) 2-Pentyne
(D) 3-Hexyne

Q16. Which type of functional group is present in compound $CH_3COOCH_3$?

(A) Ester
(B) Anhydride
(C) Acid amide
(D) Ether

Q17. The oxidation state of central metal (Mn) in the following complex $Na[Mn(CO)_5(F)_2]$ is:

(A) +2 (B) +3

(C) –1 (D) +1

> Note: SUPPLEMENTARY TABLE (Atomic No / Symbol / Mass No): 1 H 1; 2 He 4; 3 Li 7; 4 Be 9; 5 B 11; 6 C 12; 7 N 14; 8 O 16; 9 F 19; 10 Ne 20; 11 Na 23; 12 Mg 24; 13 Al 27; 14 Si 28; 15 P 31; 16 S 32; 17 Cl 35.5; 18 Ar 40; 19 K 39; 20 Ca 40. —2HA-I 2309—. Page 1 of 1.

> Note: CHEMISTRY HSSC–II. Time allowed: 2:35 Hours. Total Marks Sections B and C: 68. NOTE: Answer any fourteen parts from Section 'B' and any two questions from Section 'C'. Write your answers neatly and legibly.

## Q. SECTION – B (Marks 42)

Q. 2 Attempt any FOURTEEN parts. All parts carry equal marks. (14 x 3 = 42)

(i) How can the following conversions be carried out? a. Nitro benzene → Aniline b. Acetamide → Methyl amine.

(ii) Differentiate between 'Ethanol', '2-Prpanol' and '2-Methyl 2-Propanol' on the basis of Lucas test.

(iii) Write mechanism of the reaction when two molecules of Formaldehyde react with each other in the presence of NaOH.

(iv) Write the preparation of Ethanol and 2-Propanol from Grignard reagent.

(v) How is Nylon 6,6 prepared by condensation polymerization? Write its equations.

(vi) Enlist the raw materials used for the preparation of lipstick.

(vii) Briefly explain the trends in solubility of the hydroxides and sulphates of group II elements.

(viii) Write any three advantages of atomic emission spectroscopy.

(ix) Identify any three types of electronic transitions in Formaldehyde (H–CHO) when it is subjected to ultraviolet radiations.

(x) How can Ethanol and Acetone be identified by IR spectrum?

(xi) Write chemical equations for the preparation of Acetone by hydration of a suitable alkyne.

(xii) Briefly describe two causes that explain the inert nature of alkanes towards polar reagents.

(xiii) Predict the major product in the following reactions: a. Nitrobenzene —$Cl_2/FeCl_3$→ ? b. Aniline —$HNO_2/H_2SO_4$→ ? (Note: structural diagrams of nitrobenzene and aniline are given.)

(xiv) What is homologous series? Write its two characteristics.

(xv) Write names of the following coordination compounds. a. $Na_4[Mn(CN)_6]$ b. $Cr[(NO_3)_3(NH_3)_3]$ c. $[Ni(CO)_4(F)_2]SO_4$.

(xvi) Why $Zn^{+2}$ complexes are colorless? Explain briefly with the help of d-d transitions phenomenon.

(xvii) Why is ionization energy of nitrogen greater than that of oxygen? Although the ionization energy increases along the period.

(xviii) Write chemical reactions to show the amphoteric nature of $Al_2O_3$.

(xix) Why $K_2CO_3$ is more stable than $Li_2CO_3$? Support your answer in terms of the polarizing ability of positive ion.

(xx) Why is $CCl_4$ insoluble and $SiCl_4$ soluble in water? Explain your answer by writing mechanism for hydrolysis of group IV halides.

## Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. Write mechanisms for nitration and alkylation of Benzene. b. What is Global warming? Write three adverse effects of each 'global warming' and 'acid rain' on environment.

Q. 4 a. What is esterification? Write mechanism for the acid catalyzed reaction between methanol and acetic acid. b. Briefly explain the effect of: i. Temperature ii. pH and iii. Substrate concentration on enzyme activity along with general graphical representation for each factor.

Q. 5 a. Differentiate beryllium (Be) from other members of its group in terms of its behaviors of: i. Nitrides ii. Carbides iii. Oxides iv. Reaction with alkalies v. Reaction with Hydrogen vi. Water of crystallization and its vii. Hardness. b. Explain three differences using 'Kinetic evidence', 'Stereochemical evidence' and 'Structure of substrate' between $SN_2$ and $SN_1$ reactions.

> Note: SUPPLEMENTARY TABLE (Atomic No / Symbol / Mass No): 1 H 1; 2 He 4; 3 Li 7; 4 Be 9; 5 B 11; 6 C 12; 7 N 14; 8 O 16; 9 F 19; 10 Ne 20; 11 Na 23; 12 Mg 24; 13 Al 27; 14 Si 28; 15 P 31; 16 S 32; 17 Cl 35.5; 18 Ar 40; 19 K 39; 20 Ca 40. —2HA-I 2307—

# VERSION D

> Note: CHEMISTRY HSSC-II. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 8091. Fill the relevant bubble against each question according to curriculum:

Q1. Carboxylic acid reacts with metal to form salts with the evolution of:

(A) $CO_2$ (B) $H_2$

(C) CO (D) $CH_4$

Q2. Reaction of Triglycerides with alkalies is called:

(A) Hydration (B) Saponification

(C) Hardening (D) Dehydration

Q3. Which of the following is correct formula of Ziegler catalyst?

(A) $R_3Al + ZnCl_4$ (B) $R_2Al + TiCl_4$

(C) $R_3Al + TiCl_4$ (D) $RAl + TiCl_4$

Q4. Oxidizing smog consists of high concentration of:

(A) $O_3$ (B) $SO_2$

(C) $NO_2$ (D) $Cl_2$

Q5. Which region of electromagnetic spectrum is involved in bonds vibrations?

(A) Ultraviolet (B) Infra red

(C) Radio waves (D) Microwaves

Q6. The characteristic signal of Carbonyl Carbon of acetone in I.R spectrum will be detected at:

(A) $3200cm^{-1}$ (B) $1200cm^{-1}$

(C) $800cm^{-1}$ (D) $1720cm^{-1}$

Q7. Which of the following aldehydes will be most reactive towards Nucleophiles?

(A) Benzaldehyde (B) Acetaldehyde

(C) Formaldehyde (D) Propanaldehyde

Q8. Which of the following has highest ionization energy?

(A) Al (B) Si

(C) P (D) S

Q9. Which of the following alcohols will produce least stable alkoxide ion when O–H bond breaks?

(A) Methanol (B) Ethanol

(C) 2-Propanol (D) 2-Methyl 2-Propanol

Q10. 2,3-dimethyl-2-butene on reaction with $O_3/H_2O$ yields two molecules of:

(A) Acetaldehyde (B) Acetone

(C) Formaldehyde (D) Benzaldehyde

Q11. Which of the following alcohols cannot be synthesized when Grignard reagent reacts with aldehydes and ketones?

(A) Ethanol
(B) Isobutyl alcohol
(C) Propanol
(D) Methanol

Q12. No. of chiral centres in lactic acid COOH(CH)OH / $CH_3$ are:

(A) Two
(B) One
(C) Three
(D) Four

Q13. The general formula of cycloalkanes resembles with:

(A) Alkenes
(B) Alkynes
(C) Alkanes
(D) Benzene

Q14. The only alkaline earth metal which releases $H_2$ gas when reacted with alkalies is:

(A) Be
(B) Mg
(C) Ca
(D) Sr

Q15. Which of the following is most Paramagnetic?

(A) $Fe^{3+}$
(B) $Sc^{3+}$
(C) $Ti^{4+}$
(D) $V^{+5}$

Q16. Coordination sphere in $[Cn(en)_3]SO_4$ is:

(A) Neutral
(B) Positively charged
(C) Negatively Charged
(D) Amphoteric

Q17. Which of the following hydroxides of group II elements is least soluble in water?

(A) $Ca(OH)_2$
(B) $Mg(OH)_2$
(C) $Be(OH)_2$
(D) $Sr(OH)_2$

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## Q. SECTION – B (Marks 42)

Q. 2 Attempt any FOURTEEN parts. All parts carry equal marks. (14 x 3 = 42)

(i) Why group II carbonates are thermally unstable as compared to group I carbonates?

(ii) Why salts of $Be^{+2}$ ion cannot have more than four molecules of water of crystallization?

(iii) Write chemical reactions to show the amphoteric nature of BeO.

(iv) Why $CO_2$ is gas while $SiO_2$ is solid at room temperature? Draw structural diagrams of $CO_2$ and $SiO_2$.

(v) Describe with justification of the anomalous trends in the ionization potential of $3^{rd}$ period elements.

(vi) Write IUPAC names of following coordination compounds. a. $(NH_4)_2[Pt(Cl)_6]$ b. $[CO(F)_6(Br)_2]SO_4$ c. $[Ni(CO)_3]$.

(vii) How $V_2O_5$ acts as a catalyst in contact process in the given reaction? $SO_2 + 1/2O_2$ — $V_2O_5$→ $SO_4$.

(viii) What is meant by a. Partial Synthesis b. Total Synthesis of organic compounds in lab?

(ix) Why $Pb^{+4}$ compounds are less stable and act as an oxidizing agent? Explain with help of inert pair effect.

(x) Why bond enthalpy of F–F is less than Cl–Cl? Although size of Chlorine is bigger than that of Fluorine.

(xi) Differentiate between metamerism and tautomerism with a suitable example of each.

(xii) Which two factors determine the reactivity of alkyl halides? Write correct order of reactivity.

(xiii) Write product and mechanism for the following reaction. $CH_2{=}CH_2 + Br_2 \rightarrow ?$

(xiv) Differentiate between 1-butyne and 2-butyne by ozonolysis.

(xv) Which of the following is more acidic? Give reason. a. 2-Methyl Phenol b. 2-Nitro Phenol.

(xvi) How can Ethanol and Methanol be differentiated by iodoform test? Write chemical equation for the test.

(xvii) Briefly explain any three sources by which raw materials are obtained for chemical industries.

(xviii) What is spectroscopy? Write its principle.

(xix) Write three applications of atomic absorption spectroscopy.

(xx) Differentiate between qualitative and quantitative analysis with suitable example of each.

## Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. What are carbohydrates? Briefly describe i. Monosaccharides, ii. Disaccharides and iii. Polysaccharides with suitable examples. b. Write mechanism for the formation of an ester by a reaction between ethanol and acetic acid catalyzed by an acid.

Q. 4 a. What is sulphonation? Write mechanism for sulphonation of Benzene. b. Write reactions and predict the products when Methyl Magnesium Bromide ($CH_3MgBr$) reacts with: (i) $CO_2$ (ii) $CH_3–CH_2–C(=O)–CH_3$ (iii) H–C(=O)–H.

Q. 5 a. What is acid rain? How it occurs? Write its four adverse effects on environment. b. Justify the given order of acidity with reason. Carboxylic Acid > Phenol > Alcohol.

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